

Besmala Bendif

AI Engineer & Data Scientist



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📅 2003-07-31

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👤 Profile

AI Engineer with strong experience in deep learning, computer vision, natural language processing, and data analysis. Skilled in Python, PyTorch, TensorFlow, and transformer based architectures. Experienced in building and deploying models for image processing, NLP tasks, classification, segmentation, predictive analytics, and end-to-end AI pipelines. Passionate about developing scalable AI solutions, optimizing model performance, and applying data-driven approaches to solve real-world challenges across diverse industries.

🎓 Education

Engineering & Master's Degree in Artificial Intelligence and Data Science

2021 – 2026

National Higher School of Artificial Intelligence (ENSIA), Algiers, Algeria

Baccalaureate diploma

07/2021

Graduated with **17.73/20**

Mathematics specialization

📁 Projects

AI-Powered Automated Diagnosis of Macular Diseases Using OCT Imaging

10/2025 – 06/2026

Graduation Thesis Project

- Developed a multimodal deep learning system for macular disease diagnosis from OCT scans by integrating image features with clinical metadata (age, gender, eye side). Applied Vision Transformers, and self-supervised learning for binary and multiclass classification under patient-level evaluation.
- Integrating Explainable-AI, attention maps and segmentation to enhance model interpretability.
- Developed a full desktop-based clinical system for macular disease diagnosis, including clinic database management, patient information handling, and integration of the trained AI models for automated OCT analysis and clinical decision support.

A Deep Learning Framework on Macular OCT for Automated Multi-Attribute Metadata Recovery for Large-Scale Ophthalmic Biobanks

2026

Published in Springer LNCS Conference Proceedings (CSA 2026)

Developed a deep learning framework for imputing missing patient metadata (age, sex, eye laterality) from OCT B-scans.

- Collected and curated 53,000+ images dataset from HAMA forest.
- Fine-tuned Swin Transformer achieving high-accuracy species recognition.
- Demonstrated strong expertise in transformer-based computer vision architectures.

Arabic Vocabulary Teaching with NLP10/2024 – 01/2025

Manually collected data from ~700 Arabic books, applied OCR and preprocessing, and developed models (AR-GPT5, BERT) to support Arabic-English vocabulary learning and semantic understanding.

Forest & Wheat Mapping with Noisy Labels09/2024

Algerian Space Agency Hackathon

- Designed multi-modal models using Sentinel-1/2 data for large-scale land cover mapping.
- Leveraged self-supervised learning, transfer learning, and noise-robust methods.
- Improved model generalization in low-quality and noisy annotation environments.

Reinforcement Learning for Fraud Detection in Financial Systems02/2025 – 05/2025

Implemented RL agents with DQN, A3C on financial transaction datasets to detect fraudulent activity, exploring adaptive fraudster scenarios for robust detection.

 Professional Experience

AI Engineer Intern – Algerian Space Agency09/2024 – 10/2024 | ASAL

- Built semantic segmentation and object detection models (U-Net, DeepLabV3, Faster R-CNN, YOLO).
- Applied foundation models (CLIP, ViT) for few-shot & zero-shot remote sensing tasks.
- Worked scene classification multi-spectral satellite imagery analysis.

 Skills

Programming: Python , C++ , SQL , JavaScript.	Machine Learning/Deep Learning
Computer Vision and NLP: OpenCV , Hugging face Transformers, NLTK.	Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn.
Git, Jupyter, React.js , Figma UI/UX design.	Data Tools: NumPy, Pandas, Matplotlib, Seaborn.
Good at managing time and team work.	Good at communication skills.

 Languages

Arabic : Native, English : Advanced, French : Intermediate

 Interests

- Artificial Intelligence & Machine Learning
 - Generative AI & Large Language Models
- Computer Vision & Multimodal Models
- Natural Language Processing